

NCERT Exemplar Solutions for Class 12

Chemistry

Chapter 9 – Coordination Compounds

I. Multiple Choice Questions (Type-I)

1. Which of the following complexes formed by Cu^{2+} ions are most stable?

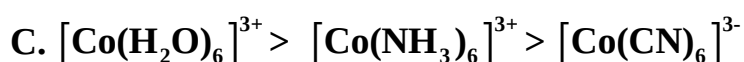
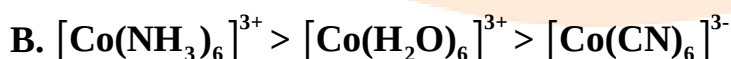
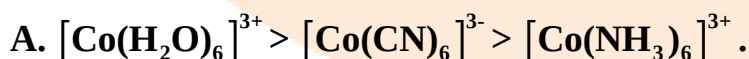


Ans: Correct options: B

The bigger the value of the stability of constant, the better the stability.

Here, $\text{Cu}^{2+} + 4\text{CN}^- \rightleftharpoons [\text{Cu}(\text{CN})_4]^{2-}$, $\log k = 27.3$ has the highest value of $\log k$ which corresponds to the highest value of k .

2. The colour of the coordination compounds depends on the crystal field splitting. What will be the correct order of absorption of wavelength of light in the visible region, for the complexes, $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{Co}(\text{CN})_6]^{3-}$, $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$.



Ans: Correct option: A

Strong field ligands have five degenerate energy levels, which means they have more energy separation than weak field ligands. That is, the strength of ligands increases, as does the splitting energy of the crystal field.

$$\text{Here, } \Delta E = \frac{hc}{\lambda}$$

$$\Delta E \propto \frac{1}{\lambda}$$

$$\lambda \propto \frac{1}{\Delta E}$$

The wavelength decreases as the energy separation rises.

3. When 0.1 mol $\text{CoCl}_3(\text{NH}_3)_5$ is treated with excess of AgNO_3 , 0.2 mol of AgCl are obtained. The conductivity of solution will correspond to

- A. 1:3 electrolyte**
- B. 1:2 electrolyte**
- C. 1:1 electrolyte**
- D. 3:1 electrolyte**

Ans: Correct option: B

When 0.1 mol $\text{CoCl}_3(\text{NH}_3)_5$ is treated with excess of AgNO_3 , In the given reaction $[\text{CoCl}_3(\text{NH}_3)_5]\text{Cl}_2$ then electrolytic solution must contain $[\text{CoCl}_3(\text{NH}_3)_5]\text{Cl}^{2+}$ and two Cl^- ions.

Hence, it is 1:2 electrolyte.

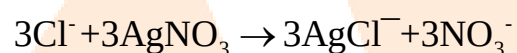
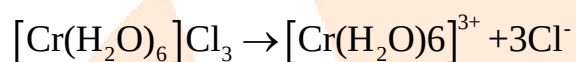
4. When 1 mol $[\text{CrCl}_3(\text{H}_2\text{O})_3] \cdot 6\text{H}_2\text{O}$ is treated with excess of AgNO_3 , 3 mol of AgCl are obtained. The formula of the complex is:

- A. $[\text{CrCl}_3(\text{H}_2\text{O})_3] \cdot 6\text{H}_2\text{O}$**
- B. $[\text{CrCl}_2(\text{H}_2\text{O})_4] \cdot \text{Cl} \cdot 2\text{H}_2\text{O}$**



Ans: Correct option: D

When 1 mol $[\text{CrCl}_3(\text{H}_2\text{O})_3]6\text{H}_2\text{O}$ is treated with excess of AgNO_3 , 3 mol of AgCl is produced, i.e., $[\text{CrCl}_3(\text{H}_2\text{O})_3]6\text{H}_2\text{O}$ is dissociated in aqueous solution and all three chloride comes in solution.



5: The correct IUPAC name of $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ is

(A) Diamminedichloridoplatinum (II)

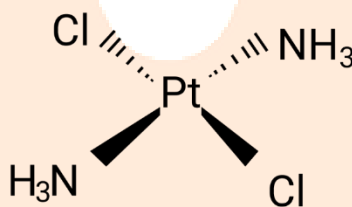
(B) Diamminedichloridoplatinum (IV)

(C) Diamminedichloridoplatinum (0)

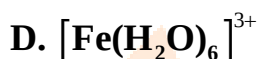
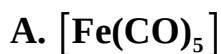
(D) Dichloridodiammineplatinum (IV)

Ans: Correct option: A

Diamminedichloridoplatinum (II) is also known as azane



6. The stabilisation of coordination compounds due to chelation is called the chelate effect. Which of the following is the most stable complex species?



Ans: Correct option: C

The chelate effect is when a chelating ligand coordinates the stabilisation of a molecule. Any ligand that binds to metal forms a ring with 5 or 6 members, which is the most stable.

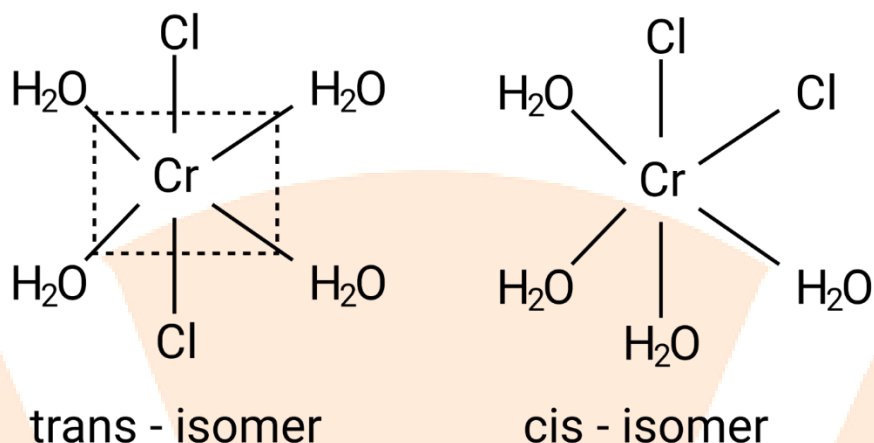
Here, CO, CN, H_2O is monodentate ligand $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ is an oxalate ligand.

7: Indicate the complex ion which shows geometrical isomerism.



Ans: Correct option: A

This is due to the complex's octahedral structure and (cis-trans) isomerism. Two Cl ligands are next to each other in trans isomerism. Although these isomers have the same structure, they are not identical.



8. The CFSE for octahedral $[\text{CoCl}_6]^{4-}$ is $18,000 \text{ cm}^{-1}$. The CFSE for tetrahedral $[\text{CoCl}_4]^{2-}$ will be

- A. $18,000 \text{ cm}^{-1}$
- B. $16,000 \text{ cm}^{-1}$
- C. $8,000 \text{ cm}^{-1}$
- D. $20,000 \text{ cm}^{-1}$

Ans: Correct option: A

CFSE for octahedral and tetrahedral complex is related as

$$\Delta_t = \frac{4}{9} \Delta_o$$

Where Δ_o = CFSE for octahedral complex

Δ_o = CFSE for tetrahedral complex

$$\Delta_o = 18000 \text{ cm}^{-1}$$

$$\Delta_t = \frac{4}{9} \times 18000 = 8000 \text{ cm}^{-1}$$

9. Due to the presence of ambidentate ligands coordination compounds show isomerism. Palladium complexes of the type $[\text{Pd}(\text{C}_6\text{H}_5)_2(\text{SCN})_2]$ and $[\text{Pd}(\text{C}_6\text{H}_5)_2(\text{NCS})_2]$ are

- (A) linkage isomers**
- (B) coordination isomers**
- (C) ionisation isomers**
- (D) geometrical isomers**

Ans: Correct option: (A)

Linkage isomerism arises in a coordination compound containing ambidentate ligand.

A simple example is by complexes containing the thiocyanate ligand, NCS^- , which may bind through the nitrogen to give $\text{M}-\text{NCS}$ or through sulphur to give $\text{M}-\text{SCN}$.

10. The compounds $[\text{Co}(\text{SO}_4)(\text{NH}_3)_5]\text{Br}$ and $[\text{Co}(\text{SO}_4)(\text{NH}_3)_5]\text{Cl}$ represent

- A. linkage isomerism**
- B. ionisation isomerism**
- C. coordination isomerism**
- D. no isomerism**

Ans: Correct option: D

Two or more compounds with the same formula but distinct structures and characteristics are known as isomers.

So, in this case compounds $[\text{Co}(\text{SO}_4)(\text{NH}_3)_5]\text{Br}$ and $[\text{Co}(\text{SO}_4)(\text{NH}_3)_5]\text{Cl}$ show no isomerism

11. A chelating agent has two or more than two donor atoms to bind to a single metal ion. Which of the following is not a chelating agent?

- A. thiosulphato**
- B. oxalato**

C glycinate

D. ethane-1,2-diamine

Ans: Correct option: A

A chelating agent binds to a single metal ion with two or more donor atoms. Oxalato, Glycinato, and ethane-1,2-diamine all have two donor oxygen atoms. The gland Thiosulpato is ambidentate.

12. Which of the following species is not expected to be a ligand?

A. NO

B. NH^{4+}

C. $\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2$

D. CO

Ans: Correct option: B

Ligands are neutral ions that form a coordination complex with the central metal atom.

NH^{4+} does not have a lone pair of electrons to give.

13. What kind of isomerism exists between $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ (violet) and $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ (greyish-green)?

A. linkage isomerism

B. solvate isomerism

C. ionisation isomerism

D coordination isomerism

Ans: Correct option: B

Hydrate isomerism is isomerism in which water is used as a solvent.

Coordination compounds with the same composition but varying ligand connectivity are known as linkage isomers.

Solvate isomerism has the same composition as free solvent but distinct solvent ligand molecules.

Except for the ligand that swaps places with the anion, ionisation isomers are identical isomers.

Coordination isomers are coordination compounds with distinct metal and ligand compositions.

14. IUPAC name of $[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NO}_2)]$ is

- A. Platinum diaminechloronitrite**
- B. Chloronitrito-N-ammineplatinum (II)**
- C Diamminechloridonitrito-N-platinum (II)**
- D Diamminechloronitrito-N-platinate (II)**

Ans: Correct option: C

$[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NO}_2)]$ is a neutral molecule. Ligands are arranged alphabetically.

So, NH_3 Diammine, chloride and the embedded linkage is nitrogen and last is the metal name.

II. Multiple Choice Questions (Type-II)

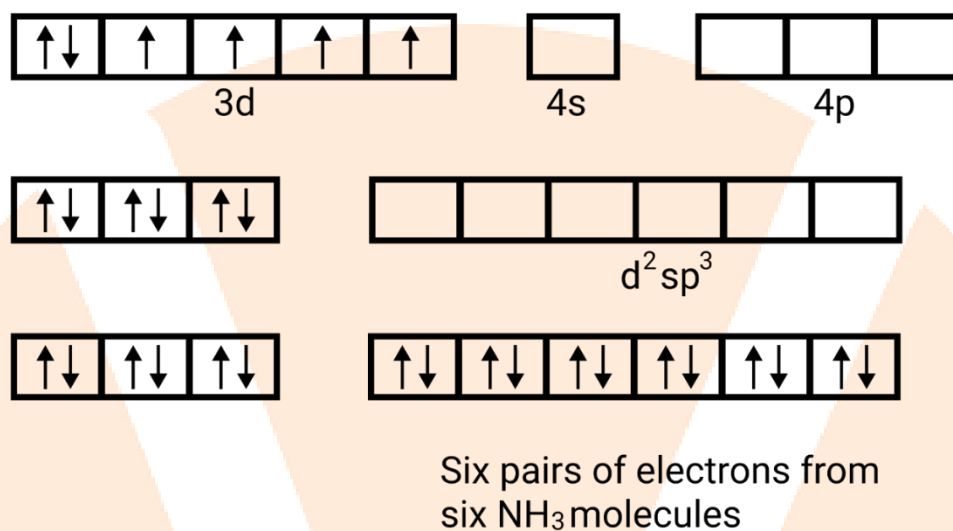
Note: In the following questions two or more options may be correct.

15. Atomic numbers of Mn, Fe, and Co are 25, 26 and 27 respectively. Which of the following inner orbital octahedral complex ions are diamagnetic?

- A. $[\text{Co}(\text{NH}_3)_6]^{3+}$**
- B. $[\text{Mn}(\text{CN})_6]^{3-}$**
- C. $[\text{Fe}(\text{CN})_6]^{4-}$**
- D. $[\text{Fe}(\text{CN})_6]^{3-}$**

Ans: Correct option: A and C

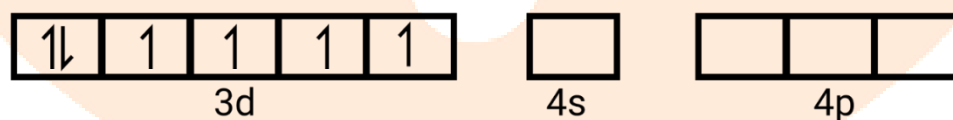
(i) Orbitals of Co^{3+} ion d^2sp^3 hybridised orbitals of $\text{Co}^{3+}[\text{Co}(\text{NH}_3)_6]^{3+}$ (inner orbital or low spin complex)



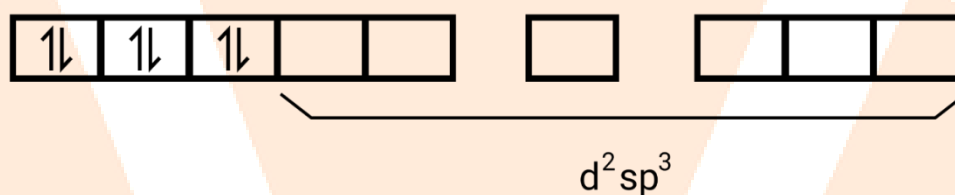
No. of unpaired electron = 0

Magnetic property = diamagnetic, due to absence of unpaired electrons

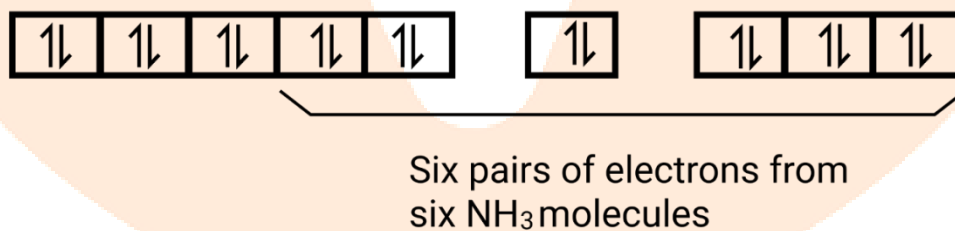
(ii) Electronic configuration is $3d^6$ orbitals of Fe^{2+} ion:



As CN^- is a strong field ligand, it causes the pairing of the unpaired 3d electrons. Since there are six ligands around the central metal ion, the most feasible hybridization d^2sp^3 hybridized orbitals of Fe^{2+} are:



6 electron pairs from CN^- ions occupy the six hybrid d^2sp^3 orbitals. Then,



Hence, the geometry of the complex is octahedral and the complex is diamagnetic (as there are no unpaired electrons.)

16. Atomic number of Mn, Fe, Co and Ni are 25, 26, 27 and 28 respectively. Which of the following outer orbital octahedral complexes have same number of unpaired electrons?



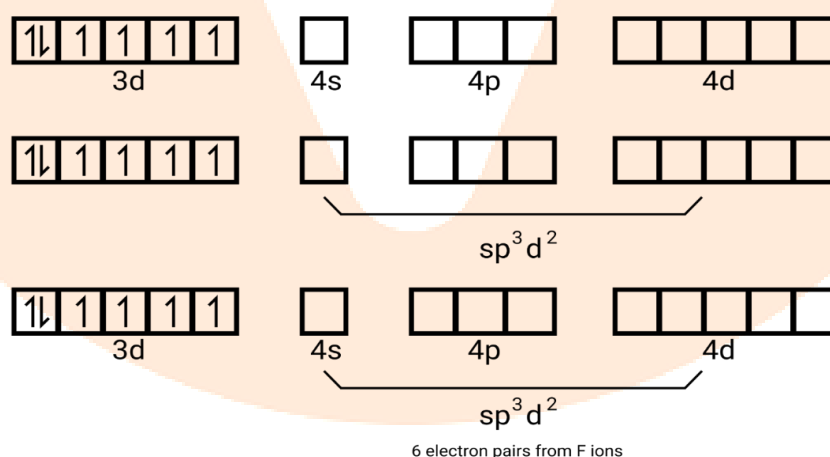
Ans: Correct option: A and C

The outer octahedral complexes are weak ligands, number of pairing electrons and high spinning potential.

$[\text{MnCl}_6]^{3-}$ - electronic configuration of Mn = $3d^5, 4s^2$. Shifting the valence electrons become $\text{Mn}^{3+}, 3d^4$ and number of unpaired electrons is 4.

$[\text{FeF}_6]^{3-}$ - electronic configuration of Fe = $3d^6, 4s^2$. Shifting the valence electrons become $\text{Fe}^{3+}, 3d^5$ and the number of unpaired electrons is 5.

$[\text{CoF}_6]^{3-}$ - electronic configuration of Co = $3d^7, 4s^2$ shifting the valence electrons become $\text{Fe}^{3+}, 3d^6$ and the number of unpaired electrons is 4.



$[\text{Ni}(\text{NH}_3)_6]^{2+}$ - electronic configuration of Ni = $3d^8, 4s^2$. Shuffling the valence electrons become $\text{Ni}^{2+}, 3d^6$ and the number of unpaired electrons is 2.

17. Which of the following options are correct for $[\text{Fe}(\text{CN})_6]^{3-}$ complex?

A. d^2sp^3 hybridisation

B. sp^3d^2 hybridisation

C. paramagnetic

D. diamagnetic

Ans: Correct option: A and C

Electronic configuration of $\text{Fe}_{(26)} = 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^6$

CN^- has a strong field ligand and electron pairing.

So, the number of unpaired electrons is 1 and paramagnetic.

18. An aqueous pink solution of cobalt (II) chloride changes to deep blue on addition of excess of HCl. This is because_____.

A. $[\text{Co}(\text{H}_2\text{O})]^{2+}$ is transformed into $[\text{CoCl}_6]^{4-}$

B. $[\text{Co}(\text{H}_2\text{O})]^{2+}$ is transformed into $[\text{CoCl}_6]^{4-}$

C. tetrahedral complexes have smaller crystal field splitting than octahedral complexes.

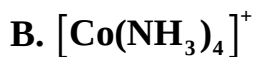
D. tetrahedral complexes have larger crystal field splitting than octahedral complex.

Ans: Correct option: B and C

$[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ is increased when excess of HCl is added. Tetrahedral complexes have smaller crystal field splitting than octahedral complexes because $\Delta_t = \frac{4}{9} \Delta_o$

19. Which of the following complexes are homoleptic?

A. $[\text{Co}(\text{NH}_3)_6]^{3+}$

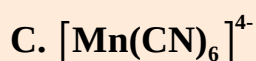


Ans: Correct options: A, B and C

A homoleptic complex is one that has only one species or group as a ligand.

Option A, B, and C each have only one ligand, but option D has two.

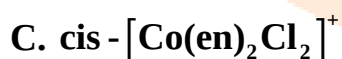
20. Which of the following complexes are heteroleptic?



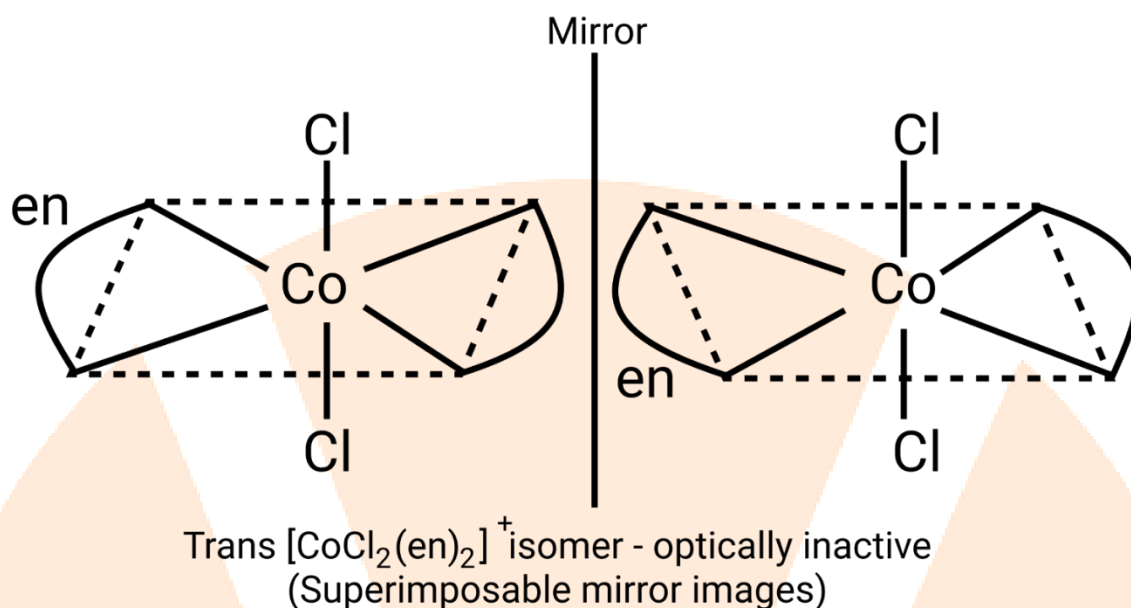
Ans: Correct option: B and D

Heteroleptic is a compound that contains many ligands of various sorts. NH_3 and Cl as a ligand or also called as donor groups, is a heteroleptic complex.

21. Identify the optically active compounds from the following.



Ans: Correct option: A and C



Optical compounds are highly imposable compounds that lack symmetry components. As a result, Option A and C are mirror reflections of each other.

22. Identify the correct statements for the behaviour of ethane-1, 2-diamine as a ligand.

- A. It is a neutral ligand.**
- B. It is a didentate ligand.**
- C. It is a chelating ligand.**
- D. It is a unidentate ligand.**

Ans: Correct options: A, B and C

Option A, B, and C describe that ethane-1, 2-diamine is a neutral ligand due to its lack of charge, a bidentate ligand due to the presence of two donor sites on one nitrogen atom, and a chelating ligand due to its ability to chelate with metal.

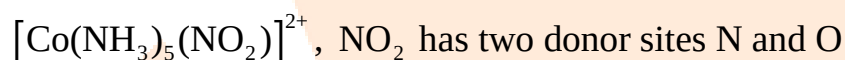
23. Which of the following complexes show linkage isomerism?

- A. $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]^{2+}$**
- B. $[\text{Co}(\text{H}_2\text{O})_5]^{3+}$**



Ans: Correct option: C

Coordination compounds with the same composition but varying ligand-metal connectivity are known as linkage isomerism.

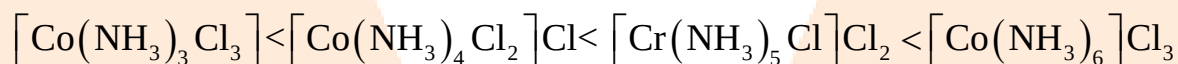


III. Short Answer Type Question

24. Arrange the following complexes in the increasing order of conductivity of their solution:



Ans: Number of ions and conductivity increases.



25. A coordination compound $\text{CrCl}_3 \cdot 4\text{H}_2\text{O}$ precipitates silver chloride when treated with silver nitrate. The molar conductance of its solution corresponds to a total of two ions. Write structural formula of the compound and name it.

Ans: The solution $\text{CrCl}_3 \cdot 4\text{H}_2\text{O}$ molar conductance shows that it contains one positive and one negative ion.

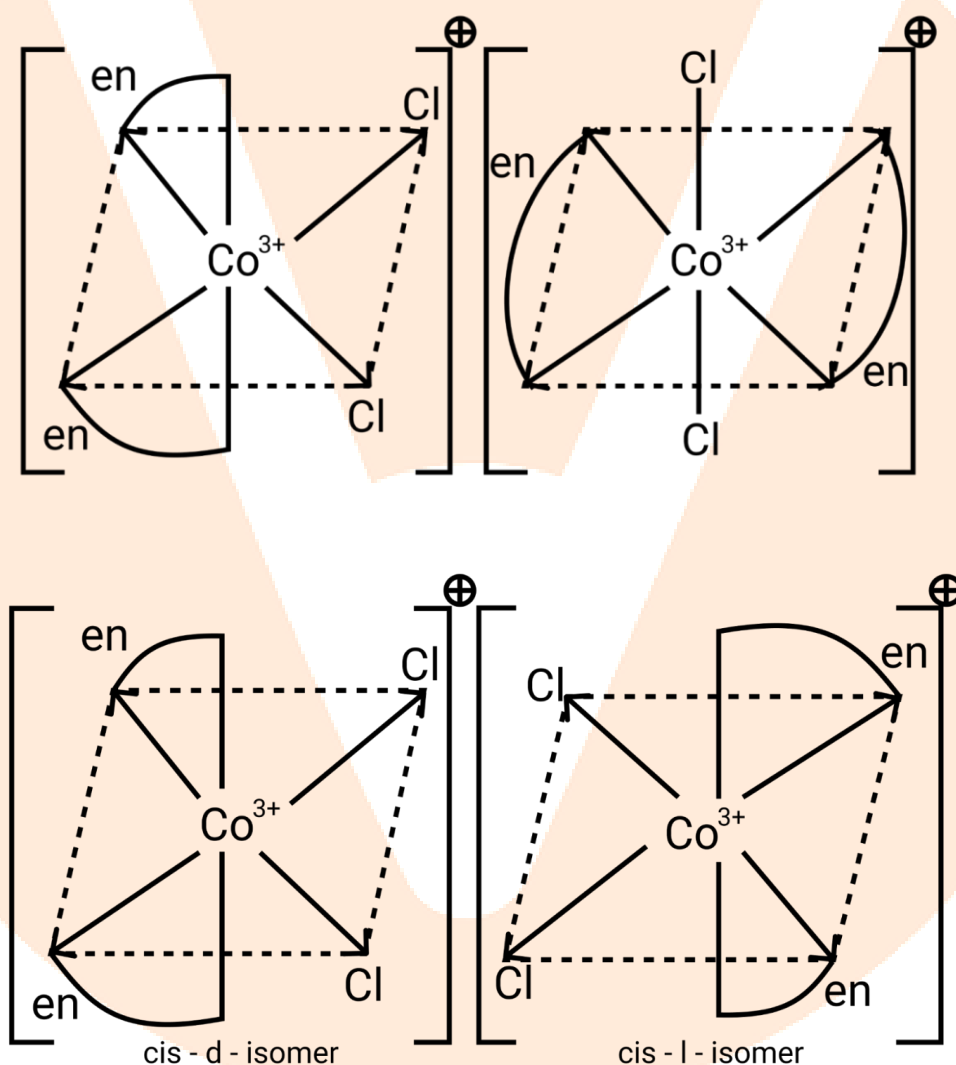
An ion is present outside the complex when silver chloride is treated with silver nitrate chloride. Outside the complex, there are two ions and one chloride.

Hence the name of the complex will be:



26. A complex of the type $[M(AA)_2X_2]^{n+}$ is known to be optically active. What does this indicate about the structure of the complex? Give one example of such a complex.

Ans: $[M(AA)_2X_2]^{n+}$ is a bidentate ligand and X is a monodentate ligand. It is an example of octahedral geometry as well as an ion. There are cis and trans isomers of this ion. Trans isomers are symmetrically active, while cis isomers are optically active.



27. Magnetic moment of $[MnCl_4]^{2-}$ is 5.92 BM. Explain giving reason.

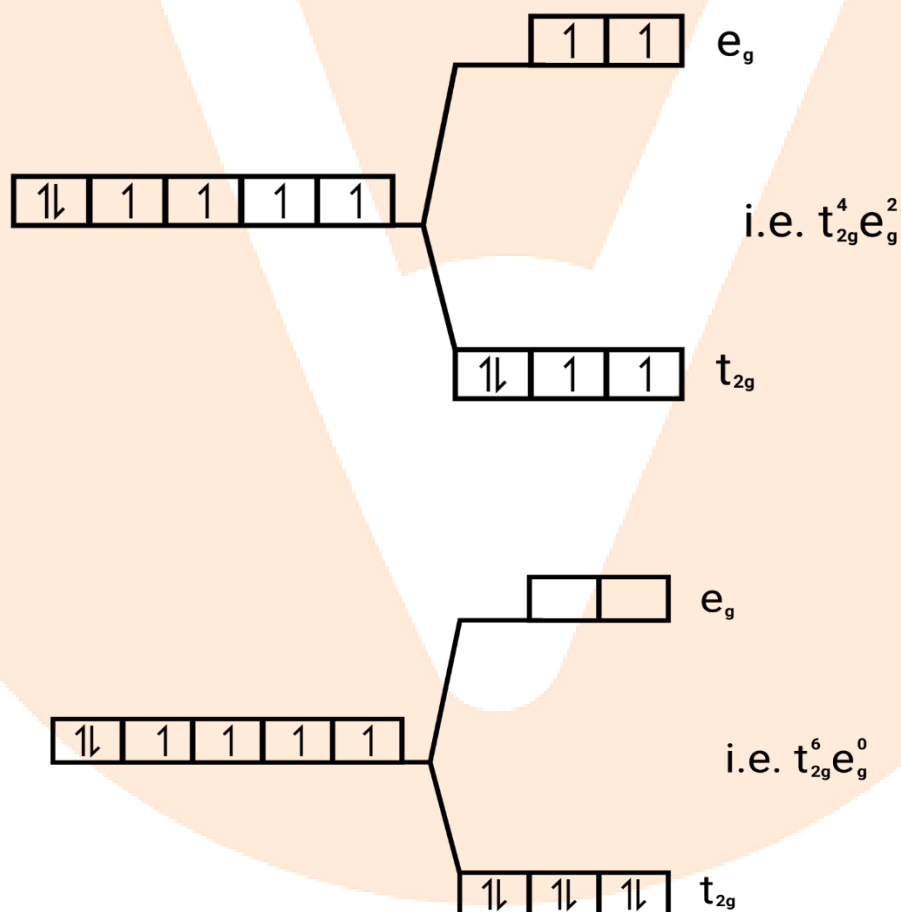
Ans: The existence of five unpaired electrons in the ion's mid-orbits Mn^{2+} equates to a magnetic moment of 5.92 BM. The number of hybridizations in this case is

sp^3 . As a result, the magnetic moment value of the tetrahedral structure complex is 5.92 BM.

28. Based on crystal field theory explain why Co(III) forms a paramagnetic octahedral complex with weak field ligands whereas it forms a diamagnetic octahedral complex with strong field ligands.

Ans: Weak field ligands, Electronic configuration of Co(III) is $t_{2g}^4 e_g^0$, has 4 paired electrons and it is paramagnetic.

Strong field ligands, Electronic configuration of is Co(III) $t_{2g}^6 e_g^0$.

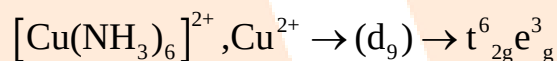
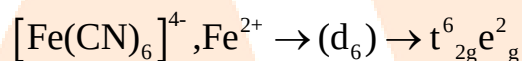
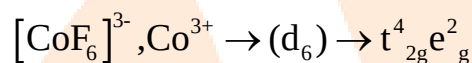


29. Why are low spin tetrahedral complexes not formed?

Ans: In a tetrahedral complex, the d-orbit splits too little compared to an octahedral complex. As a result, orbital energy alone are insufficient to couple. Low spin tetrahedral complexes do not form as a result.

30. Give the electronic configuration of the following complexes based on Crystal Field Splitting theory $[\text{CoF}_6]^{3-}$, $[\text{Fe}(\text{CN})_6]^{4-}$ and $[\text{Cu}(\text{NH}_3)_6]^{2+}$

Ans: The ligands can be grouped in ascending order of increasing field intensity according to spectrochemical series.



31. Explain why $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ has magnetic moment value of 5.92 BM whereas $[\text{Fe}(\text{CN})_6]^{3-}$ has a value of only 1.74 BM.

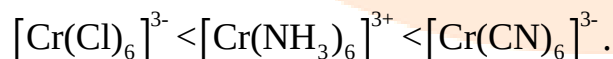
Ans: CN^- is a strong ligand and $[\text{Fe}(\text{CN})_6]^{3-}$ has one paired electron

H_2O is a weak ligand and $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ has five paired electrons.

32. Arrange following complex ions in increasing order of crystal field splitting energy (Δ_o)



Ans: The ascending value order is as follows:



The spectrochemical series includes this order.

33. Why do compounds having similar geometry have different magnetic moments?

Ans: Due to the presence of weak and strong field ligands in complexes, compounds with comparable geometry have distinct magnetic moments. The magnetic moment decreases when the CFSE increases, and vice versa.

Example: $[\text{Fe}(\text{CN})_6]^{3-}$ - Fe^{3+} , $3d^5$, CN^- (strong field ligand, pairs electron)

$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ - Fe^{3+} , $3d^5$, H_2O (weak field ligand, does not pair)

34. $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is blue in colour while CuSO_4 is colourless. Why?

Ans: Water acts as a ligand in $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ of the cases. Splitting occurs when a ligand assaults a central atom because the ligand is negatively charged and the core atom is positively charged. In CuSO_4 water is absent and Crystal field splitting is not possible and colour is formed.

35. Name the type of isomerism when ambidentate ligands are attached to central metal ion. Give two examples of ambidentate ligands.

Ans: Ambidentate ligands are ligands which have two donating sites. Coordinating compounds containing ambidentate ligands show linkage isomerism due to two different binding positions. Linkage isomerism have same ligand and geometry attached to a central metal ion by different donating sites

Example NO_2 (Nitrogen is a donor and oxygen are also a donor but they both are different elements)

Example for Linkage isomerism is $\text{Fe} \cdot \text{NO}_2$. Nitrogen can be linked to Fe, and oxygen can be related to Fe as well.

IV. Matching Type

Note: In the following questions match the items given in Columns I and II

36. Match the complex ions given in Column I with the colours given in Column II and assign the correct code

Column I (Complex ion)	Column II (Colour)
A. $[\text{Co}(\text{NH}_3)_6]^{3+}$	1. Violet

B. $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$	2. Green
C. $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$	3. Pale blue
D. $[\text{Ni}(\text{H}_2\text{O})_4(\text{en})]^{2+} (\text{aq})$	4. Yellowish orange
	5. Blue

Code

- (i) A (1) B (2) C (4) D (5)
(ii) A (4) B (3) C (2) D (1)
(iii) A (3) B (2) C (4) D (1)
(iv) A (4) B (1) C (2) D (3)

Ans: Correct option: (ii)

37. Match the coordination compounds given in Column I with the central metal atoms given in Column II and assign the correct code

Column I (Coordination Compound)	Column II (Central metal atom)
A. Chlorophyll	1. Rhodium
B. Blood pigment	2. Cobalt
C. Wilkinson catalyst	3. calcium
D. Vitamin B12	4. iron
	5. magnesium

Code

- (i) A (5) B (4) C (1) D (2)
(ii) A (3) B (4) C (5) D (1)
(iii) A (4) B (3) C (2) D (1)
(iv) A (3) B (4) C (1) D (2)

Ans: Correct option: (i)

38. Match the complex ions given in Column I with the hybridisation and number of unpaired electrons given in Column II and assign the correct code

Column I (Complex ion)	Column II (Hybridization, number of unpaired electrons)
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A. $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$	1. $\text{dsp}^2, 1$
B. $[\text{Co}(\text{CN})_4]^{2-}$	2. $\text{sp}^3\text{d}^2, 5$
C. $[\text{Ni}(\text{NH}_3)_6]^{2+}$	3. $\text{d}^2\text{sp}^3, 3$
D. $[\text{MnF}_6]^{4-}$	4. $\text{sp}^3, 4$
	5. $\text{sp}^3\text{d}^2, 2$

Code

- (i) A (3) B (1) C (5) D (2)
 (ii) A (4) B (3) C (2) D (1)
 (iii) A (3) B (2) C (4) D (1)
 (iv) A (4) B (1) C (2) D (3)

Ans: Correct option: (ii)

39. Match the complex species given in Column I with the possible isomerism given in Column II and assign the correct code

Column I (Complex species)	Column II (Isomerism)
A. $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$	1. Optical
B. $\text{cis} - [\text{Co}(\text{en})_2\text{Cl}_2]^+$	2. Ionisation
C. $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$	3. Coordination
D. $[\text{Co}(\text{NH}_3)_5][\text{Cr}(\text{CN})_6]$	4. geometrical
	5. linkage

Code:

- (i) A (1) B (2) C (4) D (5)
 (ii) A (4) B (3) C (2) D (1)
 (iii) A (4) B (1) C (5) D (3)
 (iv) A (4) B (1) C (2) D (3)

Ans: Correct option: (iv)

40. Match the compounds given in Column I with the oxidation state of cobalt present in it (given in Column II) and assign the correct code.

Column I (Compound)	Column II (Oxidation state of Co)
A. $[\text{Co}(\text{NCS})(\text{NH}_3)_5(\text{SO}_3)]$	1. + 4
B. $[\text{Co}(\text{NH}_3)_5 4\text{Cl}_2]\text{SO}_4$	2. 0
C. $\text{Na}_4[\text{Co}(\text{S}_2\text{O}_3)_3]$	3. +1
D. $[\text{Co}_2(\text{CO})_8]$	4. +2
	5. +3

Code

(i) A (1) B (2) C (4) D (5)

(ii) A (4) B (3) C (2) D (1)

(iii) A (5) B (1) C (4) D (2)

(iv) A (4) B (1) C (2) D (3)

Ans: Correct option: (i)

V. Assertion and Reason Type

Note: In the following questions a statement of assertion followed by a statement of reason is given.

Choose the correct answer out of the following choices.

A. Assertion and reason both are true, reason is correct of assertion.

B. Assertion and reason both are true, but reason is not the correct of assertion.

C. Assertion is true, reason is false.

D. Assertion is false, reason is true

41. Assertion: Toxic metal ions are removed by the chelating ligands.

Reason: Chelate complexes tend to be more stable.

Ans: Correct option: A

Toxic metal ions are heavy metal ions that are eliminated using chelating ligands in the EDTA (ethylenediamine tetra acetic acid) process, after which the metal forms a stable compound. The chelating ligand becomes stable after forming a ring. Later, the poisonous substance is removed from the body.

42. Assertion : $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_2$ and $[\text{Fe}(\text{H}_2\text{O})_6]\text{Cl}_2$ are reducing in nature.

Reason: Unpaired electrons are present in their d-orbitals.

Ans: Correct option: B

$[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_2$ and $[\text{Fe}(\text{H}_2\text{O})_6]\text{Cl}_2$ Due to the creation of a more stable complex ion after obtaining ion, $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_2$ and $[\text{Fe}(\text{H}_2\text{O})_6]\text{Cl}_2$ are reduced in nature. Both compounds have a coordination of 6 and so form an octahedral complex. The hybridisation state of is 3 and the oxidation state of is 2+. Hybridization is $3d^4$ and $3d^6$ is 2+. Both compounds have unpaired electrons and have weak field ligands. In both compounds, removing electrons is simple.

43. Assertion: Linkage isomerism arises in coordination compounds containing ambidentate ligand.

Reason: Ambidentate ligand has two different donor atoms.

Ans: Correct option: A

The term "linkage isomerism" refers to two molecules that have similar ligands but differ in the donating site of the ligand. Ambidentate is made up of two different types of donor atoms.

Example $[\text{Fe}(\text{NO}_2)(\text{H}_2\text{O})_5]\text{Cl}_2$. NO_2 is Ambidentate ligand as N and O_2 are donors.

As a result, linkage isomerism occurs in coordinations with ambidentate ligands. Reason and Assertion are both correct.

44. Assertion: Complexes of MX_6 and MX_5L type (X and L are unidentate) do not show geometrical isomerism.

Reason: Geometrical isomerism is not shown by complexes of coordination number 6.

Ans: Correct option: B

From the Cis and Trans forms, geometrical isomerism forms compounds. All of the valences in the MX_6 ligand are the same, resulting in an identical molecule. L is a distinct ligand in MX_5L , and it can be replaced with any other valency in the structure to create an identical molecule; the molecule does not change.

Because of the presence of symmetry, which is an essential prerequisite for revealing geometrical isomerism, complexes do not show geometrical isomerism.

45. Assertion: $[\text{Fe}(\text{CN})_6]^{3-}$ ion shows magnetic moment corresponding to two unpaired electrons. **Reason:** Because it has d^2sp^3 type hybridisation.

Ans: Correct option: D

$[\text{Fe}(\text{CN})_6]^{3-}$ ion shows magnetic moment corresponding to one unpaired electron. (CN) is a ligand with a strong field that couples electrons, resulting in hybridisation. d^2sp^3 and it has one unpaired electron, it has the magnetic moment of one paired electron.

$$\begin{aligned}\mu &= \sqrt{n(n+2)} \\ &= \sqrt{1(1+2)} \\ &= \sqrt{3} = 1.73 \text{ BM}\end{aligned}$$

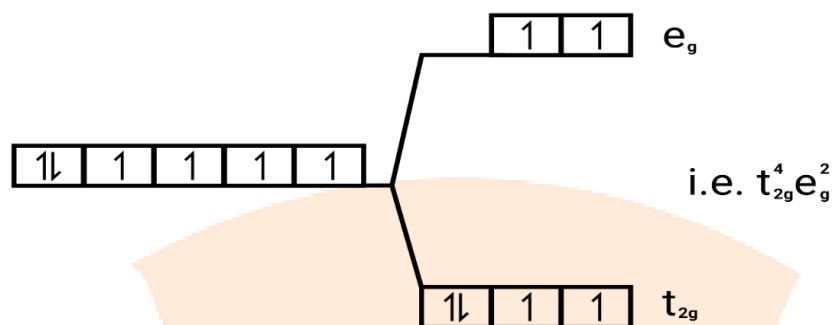
VI. Long Answer Type

46. Using crystal field theory, draw energy level diagram, write electronic configuration of the central metal atom/ion and determine the magnetic moment value in the following:

(i) $[\text{CoF}_6]^{3-}$, $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$, $[\text{Co}(\text{CN})_6]^{3-}$

Ans: Electronic configuration: $\text{Co}^{3+} = [\text{Ar}]3d^6$

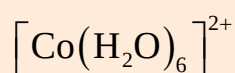
Energy level diagram:



Magnetic moment:

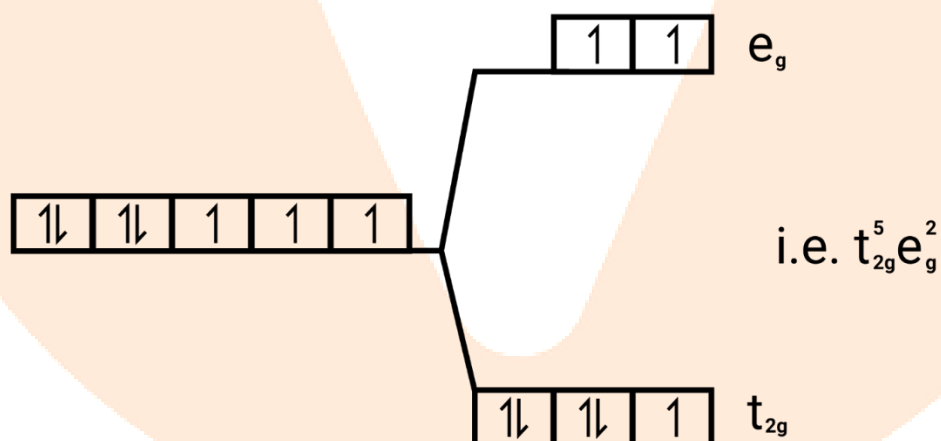
Number of unpaired electrons (n) = 4

$$\text{Magnetic moment} = \sqrt{n(n+2)} = \sqrt{4(4+2)} = \sqrt{24} = 4.9 \text{ BM}$$



Electronic configuration: $\text{Co}^{2+} = [\text{Ar}]3d^7$

Energy level diagram:

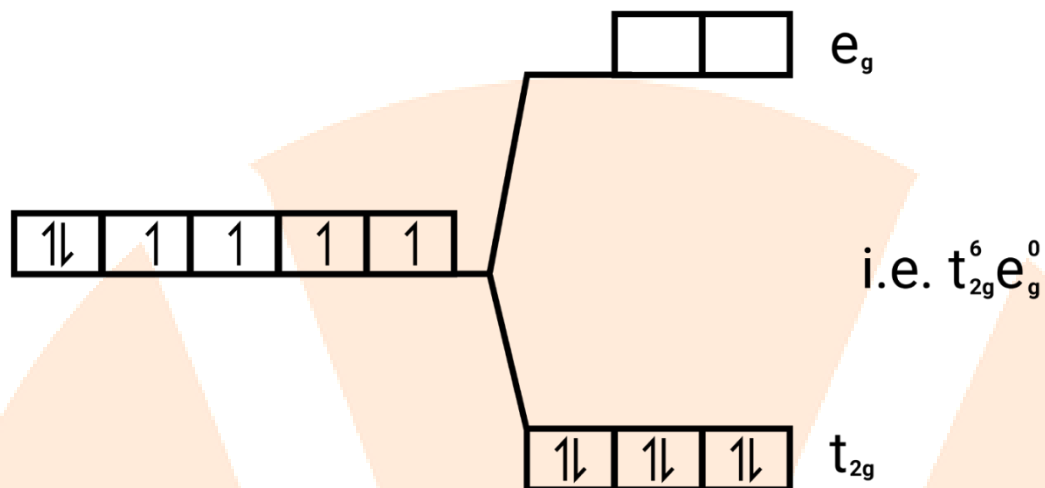


Magnetic moment: Since, number of unpaired electrons (n) = 3, therefore
magnetic moment = $\sqrt{3(3+2)} = \sqrt{15} = 3.87 \text{ BM}$



Electronic configuration: $[\text{Ar}]\text{Co}^{3+} = 3d^6$

Energy level diagram:

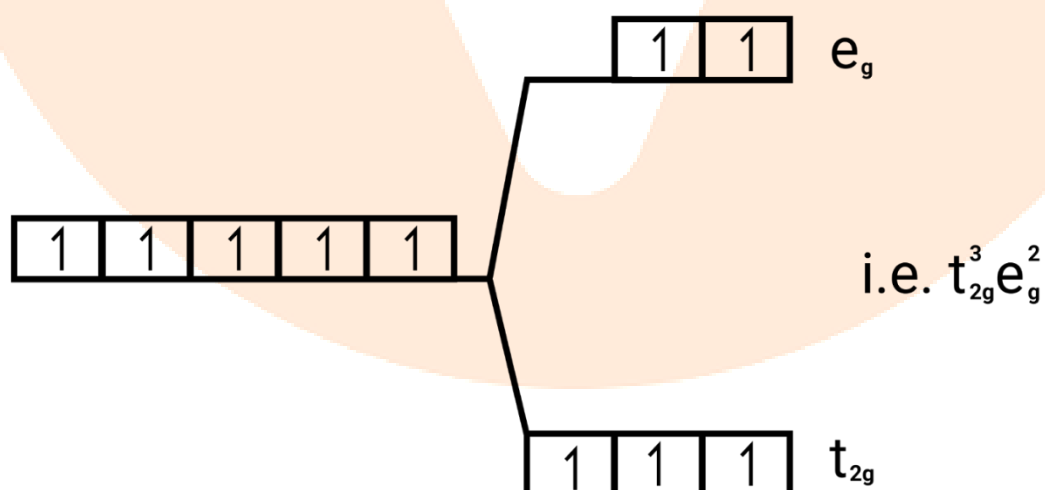


(ii) $[\text{FeF}_6]^{3-}$, $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$, $[\text{Fe}(\text{CN})_6]^{4-}$

Ans: $[\text{FeF}_6]^{3-}$

Electronic configuration: $\text{Fe}^{3+} = [\text{Ar}]3d^5$

Energy level diagram:



Number of unpaired electrons = 5 $\sqrt{5(5+2)} = \sqrt{35} = 5.95\text{BM}$ $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$

$\text{Fe}^{2+} = [\text{Ar}]3\text{d}^6$

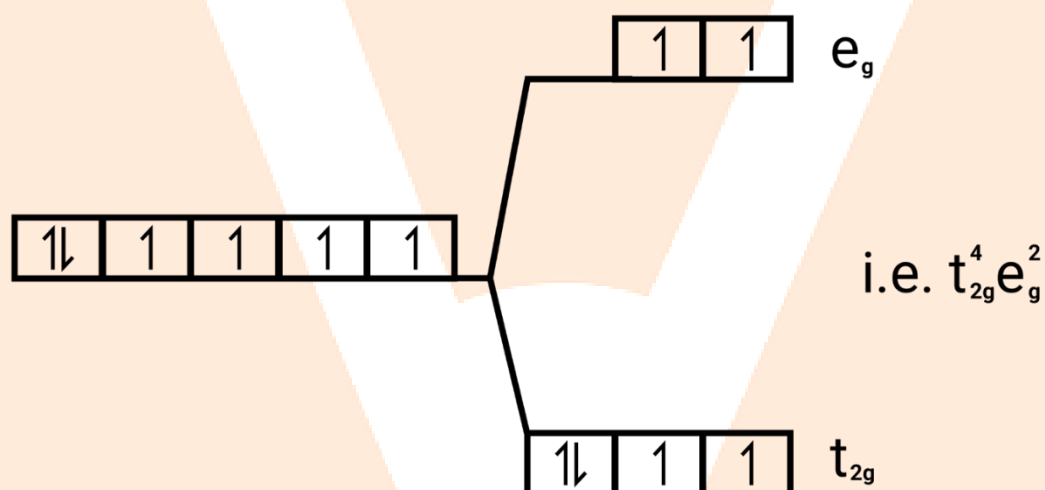
Energy level diagram:

Magnetic moment

Since, number of unpaired electrons (n) = 4, therefore magnetic moment is-

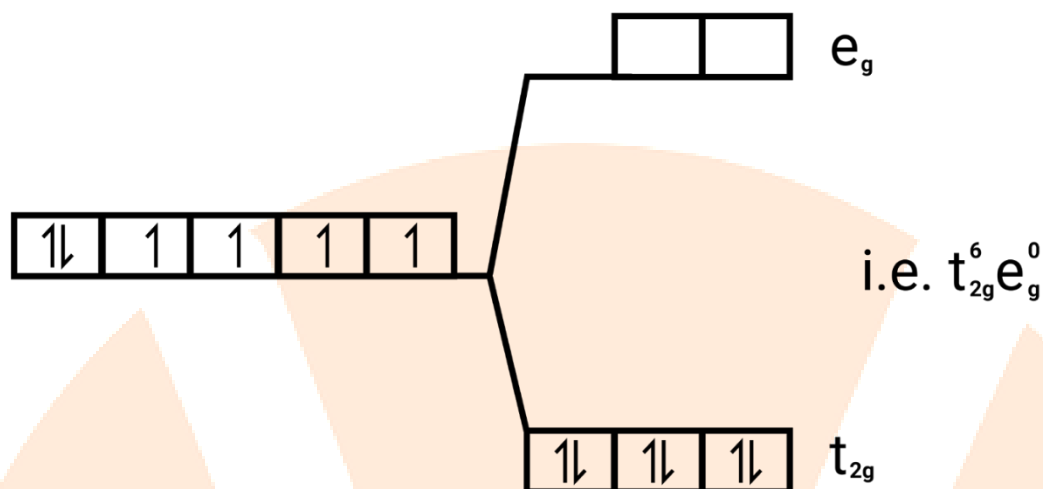
$$\sqrt{4(4+2)} = \sqrt{24} = 4.9\text{BM}$$

$[\text{Fe}(\text{CN})_6]^{4-}$



Electronic configuration: $\text{Fe}^{2+} = [\text{Ar}]3\text{d}^6$

Energy level diagram

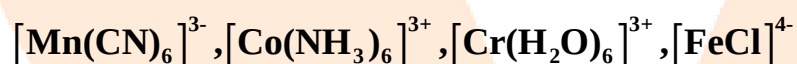


Since CN^- is strong field ligand, it forces all the electrons to get paired up
Magnetic moment.

In the absence of unpaired electrons it behaves as diamagnetic (ie. no magnetic behaviour).

So magnetic moment = zero

47. Using valence bond theory, explain the following in relation to the complexes given below:



(i) Type of hybridisation.

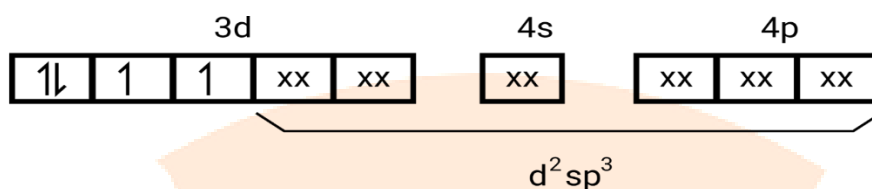
(ii) Inner or outer orbital complex.

(iii) Magnetic behaviour.

(iv) Spin only magnetic moment value

Ans: $[\text{Mn}(\text{CN})_6]^{3-}$

Electronic configuration is $\text{Mn}^{3+} = [\text{Ar}]3d^4$ hence box electronic structure



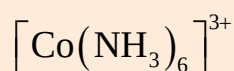
(i) Type of hybridisation d^2sp^3

(ii) Inner orbital complex

(iii) paramagnetic, due to presence of three unpaired electrons.

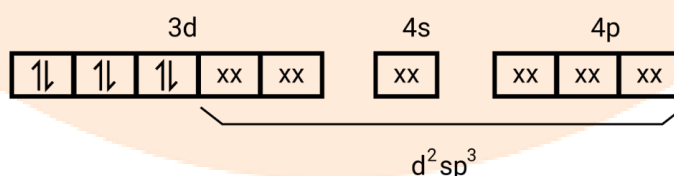
(iv) Spin only magnetic moment is calculated using the formula : $n = 2$ in this case, we get spin only magnetic moment in BM as $= \sqrt{2(2 + 2)} = \sqrt{8} =$

2.87BM



Electronic configuration of $\text{Co}^{3+} = [\text{Ar}]3d^6$

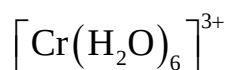
(i) Hyb As shown in the above box electronic structure the type of hybridisation is d^2sp^3



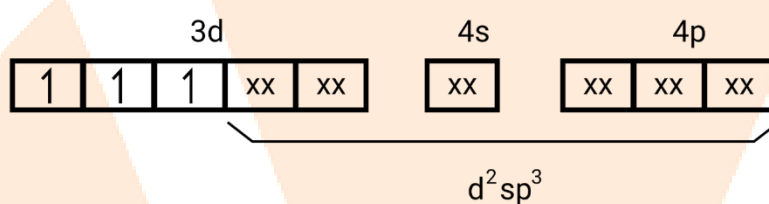
(ii) Inner orbital complex

(iii) Diamagnetic

(iv) Zero(Since no unpaired electrons are present, as depicted above)



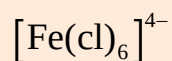
(i) type of hybridisation..... d^2sp^3



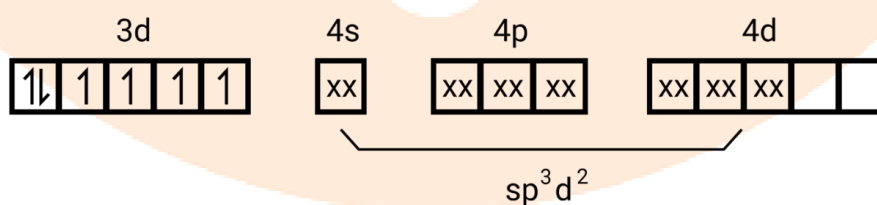
(ii) Inner orbital complex

(iii) paramagnetic

(iv) 3.87BM



electronic configuration is of $\text{Fe}^{2+} = [\text{Ar}]3d^6$



(i) type of hybridisation will be sp^3d^2

(ii) Outer orbital complex

(iii) Paramagnetic

(iv) 4.9BM

48. $\text{CoSO}_4\text{Cl} \cdot 5\text{NH}_3$ exists in two isomeric forms 'A' and 'B'. Isomer 'A' reacts with AgNO_3 to give white precipitate but does not react with BaCl_2 . Isomer 'B' gives white precipitate with BaCl_2 but does not react with AgNO_3 . Answer the following questions.

(i) Identify 'A' and 'B' and write their structural formulas.

Ans: 'A' is $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Cl}$

'B' is $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{SO}_4$

(ii) Name the type of isomerism involved.

Ans: Type of isomerism is ionisation isomerism

(iii) Give the IUPAC name of 'A' and 'B'.

Ans: IUPAC name of isomer 'A' is pentaaminesulphatocobalt(III)chloride

IUPAC name of isomer 'B' is pentaaminechlorocobalt(III)sulphate

49. What is the relationship between observed colour of the complex and the wavelength of light absorbed by the complex?

Ans: When light wavelengths from a specific part of the spectrum are absorbed by a substance, the result is a complimentary colour. When a complex absorbs a wavelength of light, it reflects a complementary colour. If a violet colour is absorbed, for example, yellow is conveyed. The CFSE value, often known as the colour definer for any complex, comes next. To determine the wavelength value in order to determine which colour absorbs the most energy.

$$\Delta E = \frac{hc}{\lambda}$$

As λ has shorter wavelength.

Low spin complexes absorb shorter wavelengths, while high spin complexes absorb longer ones. The colour theory, complex, and wavelength are all determined using this method.

50. Why are different colours observed in octahedral and tetrahedral complexes for the same metal and same ligands?

Ans: The octahedral and tetrahedral splitting up of d orbitals is different.

$$\Delta_t = \left(\frac{4}{9}\right) \Delta_o$$

$$\Delta_t < \Delta_o$$

Δ_t = CFSE in tetrahedral field

Δ_o = CFSE in octahedral field

Comparing CFSE with energy = $\frac{hc}{\lambda}$

The lower the absorption in, the higher the CFSE. The lower wavelengths are absorbed by the octahedral complex, while the rest is transmitted. It absorbs higher wavelengths in a tetrahedral complex and transmits the rest. For the same metal and ligands, the octahedral complex has a shorter wavelength than the tetrahedral complex.